Unit 2 Progress Check: MCQ Part A

1. The variables x, y, and z are integers. Which of the following algorithms can be used to swap the values of the variables so that they are ordered from least to greatest (i.e., $x \leq y \leq z$)?

Unit 2 Progress Check: MCQ Part A

(A)

Step 1:	If x is greater than y, swap the values of x and y.
Step 2:	If y is greater than z, swap the values of y and z.

(B)

Step 1:	If x is greater than z, swap the values of x and z.
Step 2:	If y is greater than z, swap the values of y and z.

(C) ✓

Step 1:	If x is greater than y, swap the values of x and y.
Step 2:	If y is greater than z, swap the values of y and z.
Step 3:	If x is greater than y, swap the values of x and y.

Unit 2 Progress Check: MCQ Part A

(D)

Step 1:	If x is greater than y, swap the values of x and y.
Step 2:	If y is greater than z, swap the values of y and z.
Step 3:	If x is greater than z, swap the values of x and z.

Answer C

Correct. Step 1 ensures that x is less than or equal to y. Step 2 ensures that z is the largest number. Step 3 ensures that x is the smallest number. After all three steps are executed, the variables will be in the desired order.

Unit 2 Progress Check: MCQ Part A

2. A company is designing a program to evaluate the strength of a user's password. The program needs to indicate if the password is "weak," "medium," or "strong" based on the following guidelines.

- A password with fewer than 8 characters is classified as "weak."
- A password with 8 or more characters is classified as "medium," where all characters are either numbers or letters.
- A password with 8 or more characters plus at least one special character (like !, @, or #) is classified as "strong."

Which of the following algorithms will correctly classify a password?

(A) Step 1: If the password has fewer than 8 characters, label it as "weak" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password has more than 8 characters, label it as "medium" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "strong."

(B) ✓ Step 1: If the password has fewer than 8 characters, label it as "weak" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password contains at least one special character, label it as "strong" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "medium."

(C) Step 1: If the password has 8 or more characters, label it as "medium" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password has more than 8 characters and contains at least one special character, label it as "strong" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "weak."

(D) Step 1: If the password has 8 or more characters and contains at least one special character, label it as "strong" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password does not contain at least one special character, label it as "medium" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "weak."

Answer B

Correct. This algorithm will yield correct results each time. The algorithm first labels all passwords shorter than 8 characters as "weak" and then skips the other steps. If a password is not labeled weak, the algorithm then considers if the password has a special character and labels those "strong", and then skips the last step. Finally, if a password was not labeled "weak" or "strong", the algorithm labels the password "medium."

Unit 2 Progress Check: MCQ Part A

3. Users of a mobile application are categorized based on the average number of hours they use the application each day, as shown in the following table.

Category	Average Hours of Use Per Day
Top User	Greater than 5
Regular User	Between 1 and 5, inclusive
Inactive User	Less than 1

Suppose n is the average number of hours a user spends using the application each day. Which of the following algorithms can be used to assign the correct categorization to a user?

(A) If n is greater than 5, categorize the user as a top user. Otherwise, if n is less than or equal to 5, categorize the user as a regular user. Otherwise, categorize the user as an inactive user.

(B) If n is greater than 5, categorize the user as a top user. Otherwise, if n is greater than or equal to 1, categorize the user as a regular user. Otherwise, categorize the user as an inactive user. ✓

(C) If n is less than 1, categorize the user as an inactive user. Otherwise, if n is greater than or equal to 1, categorize the user as a regular user. Otherwise, categorize the user as a top user.

(D) If n is greater than or equal to 1, categorize the user as a regular user. Otherwise, if n is less than 1, categorize the user as an inactive user. Otherwise, categorize the user as a top user.

Answer B

Correct. This algorithm first checks if n is greater than 5, and if so, correctly categorizes the user as a top user. If this is not the case, the algorithm then checks if n is between 1 and 5, inclusive, and if so, correctly categorizes the user as a regular user. If this is not the case, the algorithm then correctly categorizes the user as an inactive user.

Unit 2 Progress Check: MCQ Part A

4. Consider the following code segment, which is intended to print `true` only when both `r` and `v` are positive numbers. Assume that `r` and `v` have been properly declared and initialized.

```
boolean rPos = r > 0;
boolean vPos = v > 0;
System.out.print(rPos == vPos);
```

The code segment does not always work as intended.

Which of the following values for `r` and `v` will demonstrate that this code segment does **not** work as intended?

(A)

r = -10, v = 10

(B) ✓

r = -1, v = -4

(C)

r = 5, v = -2

(D)

r = 6, v = 3

Answer B

Correct. For these values, the code is intended to print `false` since variables are negative. `rPos` will evaluate to `false`, and `vPos` will evaluate to `false`. The expression `rPos == vPos` will evaluate to `true`. This is not the intention of the code.

Unit 2 Progress Check: MCQ Part A

5. In the following code segment, a and b are properly declared and initialized boolean variables.

```
boolean result = (a != b);
System.out.println(result);
```

Which of the following best describes the behavior of the code segment?

(A) It prints true if a is true and b is false, and prints false otherwise.

(B) It prints true if a and b have different values, and prints false otherwise. ✓

(C) It prints true if a and b have the same value, and prints false otherwise.

(D) It prints false for all possible values of a and b.

Answer B

Correct. The expression a != b evaluates to true when a and b are not equal. It evaluates to false when a and b are equal.

Unit 2 Progress Check: MCQ Part A

6. Consider the following expression.

(3 + 4 == 5) != (3 + 4 >= 5)

What value, if any, does the expression evaluate to?

(A) ✓
true

(B)
false

(C)
5

(D) No value; relational operators cannot be used on arithmetic expressions.

Answer A

Correct. The expression 3 + 4 == 5 evaluates to false. The expression 3 + 4 >= 5 evaluates to true. These two expressions differ in value, so (3 + 4 == 5) != (3 + 4 >= 5) evaluates to true.

Unit 2 Progress Check: MCQ Part A

7. In the following code segment, n is a properly declared and initialized int variable.

```
boolean result = false;
if (n >= 10)
{
  result = true;
}
if (n <= 50)
{
  result = true;
}
System.out.println(result);
```

Which of the following best describes the behavior of the code segment?

(A) It prints true for all possible values of n. ✓

(B) It prints false for all possible values of n.

(C) It prints true if n is between 10 and 50, inclusive, and prints false otherwise.

(D) It prints false if n is between 10 and 50, inclusive, and prints true otherwise.

Answer A

Correct. The code segment initializes result to false, then uses the two if statements to set result to true if n is greater than or equal to 10 or less than or equal to 50. All possible values of n will make at least one of these conditions true, so the code segment will always print true.

Unit 2 Progress Check: MCQ Part A

8. Consider the following code segment.

```
int quant = 20;
int unitPrice = 4;
int ship = 8;
int total;
if (quant > 10)
{
  unitPrice = 3;
}
if (quant > 20)
{
  ship = 0;
}
total = quant * unitPrice + ship;
```

What is the value of total after this code segment is executed?

(A)

20

(B)

60

(C) ✓

68

(D)

88

Answer C

Correct. The value of unitPrice is initially set to 4 and then it is changed to 3 because quant > 10 is true. The value of ship is initially set to 8 and then it is not changed because quant > 20 is false. The value of total is 20 * 3 + 8, which is equal to 68.

Unit 2 Progress Check: MCQ Part A

9. The following code segment is intended to print the total cost of a stay in a room at a particular hotel. The int variable numNights represents the length of the stay, in nights, and the double variable dailyRate represents the room base cost for each night of the stay. For stays that are longer than five nights, there is a 10% discount applied to the room base cost for all nights during the stay. In addition to the room cost, the hotel charges a $25 resort fee for each night of the stay. The resort fee is not eligible for the 10% discount for stays that are longer than five nights.

```
int numNights =    /* some initial value */ ;
double dailyRate = /* some initial value */ ;

double totalCost = numNights * dailyRate;
/* missing code */

System.out.println(totalCost);
```

Which of the following can be used to replace /* missing code */ so that the code segment works as intended?

Unit 2 Progress Check: MCQ Part A

(A)

```
if (numNights > 5)
{
  totalCost *= 0.9;
  totalCost += (numNights * 25);
}
```

(B) ✓

```
if (numNights > 5)
{
  totalCost *= 0.9;
}
totalCost += (numNights * 25);
```

(C)

```
totalCost += (numNights * 25);
if (numNights > 5)
{
  totalCost *= 0.9;
}
```

Unit 2 Progress Check: MCQ Part A

(D)

```
totalCost += 25;

if (numNights > 5)

{

  totalCost *= 0.9;

}
```

Answer B

Correct. This code segment first applies the 10% discount (by multiplying by 0.9) if the stay is longer than 5 nights, and then adds the resort fee ($25 per night) for all nights.

Unit 2 Progress Check: MCQ Part A

10. Consider the following two code segments. Assume that variables x and y have been declared as int variables and have been assigned integer values.

Code Segment I

```
int result = 0;
if (x > y)
{
  result = x - y;
  System.out.print(result);
}
else if (x < y)
{
  result = y - x;
  System.out.print(result);
}
else
{
  System.out.print(result);
}
```

Code Segment II

```
if (x < y)
{
  System.out.print(y - x);
}
else
{
  System.out.print(x - y);
}
```

Which of the following correctly compares the outputs of the two code segments?

(A) Code segment I and code segment II produce the same output for all values of x and y. ✓

(B) Code segment I and code segment II produce the same output only when x is equal to y.

(C) Code segment I and code segment II produce the same output only when x is not equal to y.

(D) Code segment I and code segment II produce the same output only when x is less than y.

Answer A

Correct. Both code segments print the absolute value of the difference between x and y. When x is greater than y, both code segments print x - y. When x is less than y, both code segments print y - x. When x is equal

Unit 2 Progress Check: MCQ Part A

to y, both segments print 0.

Unit 2 Progress Check: MCQ Part A

11. Consider the following code segment, which uses properly declared and initialized int variables x and y and the String variable result.

```
String result = "";
if (x < 5)
{
  if (y > 0)
  {
    result += "a";
  }
  else
  {
    result += "b";
  }
}
else if (x > 10)
{
  if (y < 0)
  {
    result += "c";
  }
  else if (y < 10)
  {
    result += "d";
  }
  result += "e";
}
result += "f";
```

What is the value of result after the code segment is executed if x has the value 15 and y has the value 5?

(A)

adf

(B)

d

(C) ✓

def

(D)

ef

Unit 2 Progress Check: MCQ Part A

Answer C

Correct. The value of result will contain neither "a" nor "b" because x < 5 is false. Since x > 10 is true but y < 0 is false, the value of result does not contain "c". Since x > 10 is true and y < 10 is true, the value of result contains "d". The value of result contains "e" because the code line result += "e"; is outside the inner if-else structure. The value of result contains "f" because the code line result += "f"; is outside the outer if-else structure.

Unit 2 Progress Check: MCQ Part A

12. In the following code segment, str is a properly declared and initialized String variable.

```
int result = 0;
if (str.length() > 5)
{
  if (str.indexOf("A") < 0)
  {
    result = 1;
  }
  else if (str.indexOf("B") < 0)
  {
    result = 2;
  }
}
else if (str.indexOf("A") < 0)
{
  result = 3;
}
```

Which of the following code segments assigns the same value to result as the preceding code segment for all values of str?

Unit 2 Progress Check: MCQ Part A

(A)

```
int result = 0;

if (str.indexOf("A") < 0)

{

  result = 3;

}

else if (str.indexOf("B") < 0)

{

  result = 2;

}

else if (str.length() > 5);

{

  result = 1;

}
```

Unit 2 Progress Check: MCQ Part A

(B)

```
int result = 0;
if (str.indexOf("A") < 0)
{
  if (str.length() > 5)
  {
    result = 1;
  }
  else if (str.indexOf("B") < 0)
  {
    result = 2;
  }
  else
  {
    result = 3;
  }
}
```

Unit 2 Progress Check: MCQ Part A

(C)

```
int result = 0;
if (str.indexOf("A") < 0)
{
  if(str.length() > 5)
  {
    result = 1;
  }
  else
  {
    result = 3;
  }
}
else if (str.indexOf("B") < 0)
{
  if (str.length() > 5)
  {
    result = 2;
  }
}
```

Unit 2 Progress Check: MCQ Part A

(D)

```
int result = 0;
if (str.indexOf("B") < 0)
{
  if (str.length() > 5)
  {
    result = 2;
  }
}
else if (str.indexOf("A") < 0)
{
  if (str.length() > 5)
  {
    result = 1;
  }
  else
  {
    result = 3;
  }
}
```

Answer C

Correct. The code will assign result to the correct value for all possible string values.

Unit 2 Progress Check: MCQ Part A

13. A certain device requires a passcode to be used. The passcode must meet both of the following requirements.

- The passcode must be 4 digits long.
- The passcode must not start with a zero (0) digit.

Assume that `pass` is a properly declared and initialized `int` variable.

Which of the following expressions will evaluate to `true` if `pass` is an acceptable passcode?

(A)

```
pass >= 1000 || pass < 10000
```

(B)

```
pass > 1000 || pass <= 10000
```

(C) ✓

```
pass >= 1000 && pass < 10000
```

(D)

```
pass > 1000 && pass < 10000
```

Answer C

Correct. The first expression `pass >= 1000` is false for values of `pass` with fewer than 4 digits, and the second expression `pass < 10000` is false for values of `pass` with greater than 4 digits. The `&&` (and) operator requires that both expressions be `true`.

Unit 2 Progress Check: MCQ Part A

14. A fitness application categorizes daily step counts into different activity levels. The "Active" level is defined as any step count between 10,000 and 15,000, inclusive. The following code segment is intended to determine if a user's step count falls within the "Active" category.

```
if (steps > 10000 && steps < 15000)
{
  activityLevel = "Active";
}
```

Which of the following best explains the error, if any, in this code segment?

(A) There is no error in the code segment as it correctly identifies "Active" step counts.

(B) The condition should be `if (steps >= 10000 && steps <= 15000)` to properly include 10,000 and 15,000. ✓

(C) The condition should be `if (steps > 10000 || steps < 15000)` to include the correct range of step counts.

(D) The condition should be `if (steps >= 10000 || steps <= 15000)` to include all step counts from 10,000 and above, as well as 15,000 and below.

Answer B

Correct. This solution correctly checks if `steps` is greater than or equal to 10,000 and less than or equal to 15,000, matching the "Active" category definition.

Unit 2 Progress Check: MCQ Part A

15. Consider the following code segment.

```
boolean a = true;
boolean b = true;
System.out.print((b || (!a || b)) + " ");
System.out.print(((!b || !a) && a) + " ");
System.out.println(!(a && b) && b);
```

What output is produced when this code segment is executed?

(A)

true true true

(B)

true false true

(C) ✓

true false false

(D)

false true false

Answer C

Correct. Since b is true, the first logical expression evaluates to true. The expression (!a || b) is not evaluated due to short-circuit evaluation. Since !b is false and !a is false, (!b || !a) evaluates to false, and the rest of the expression is not evaluated due to short-circuit evaluation. Since (a && b) evaluates to true, !(a && b) evaluates to false, and the third logical expression evaluates to false. The expression b is not evaluated due to short-circuit evaluation.

Unit 2 Progress Check: MCQ Part A

16. In the following code segment, x is a properly declared and initialized boolean variable and y is a properly declared and initialized int variable.

```
boolean a = !x && y > 50000;
boolean b = !(x || y <= 50000);
boolean c = !x || y <= 50000;
boolean d = !(x && y <= 50000);

if (a == b)
{
  System.out.println("First");
}
if (a == c)
{
  System.out.println("Second");
}
if (b == c)
{
  System.out.println("Third");
}
if (c == d)
{
  System.out.println("Fourth");
}
```

Which of the following is always printed as a result of executing this code segment?

(A) ✓

First

(B)

Second

(C)

Third

(D)

Fourth

Answer A

Unit 2 Progress Check: MCQ Part A

Correct. Applying De Morgan's law to the definition of b yields !x && y > 50000. This is the same as a.

17. An image on a computer is made up of many pixels. Each pixel in the image contains varying amounts of red, green, and blue. A "mainly red" pixel is defined to be one in which the percentage of red is more than 50, and the percentages of green and blue are each less than 10.

Assume that redPct, greenPct, and bluePct are int variables that have been properly declared and initialized with values representing the percentages of red, green, and blue, respectively, in a pixel.

Which of the following expressions will evaluate to true for a "mainly red" pixel?

(A)

```
(redPct <= 50) || (greenPct >= 10) || (bluePct >= 10)
```

(B)

```
(redPct > 50) && (greenPct < 10) || (bluePct < 10)
```

(C) ✓

```
!( (redPct <= 50) || (greenPct >= 10) || (bluePct >= 10) )
```

(D)

```
!(redPct > 50) && !(greenPct > 10) && !(bluePct > 10)
```

Answer C

Correct. Application of De Morgan's law yields the code segment if (redPct > 50 && greenPct < 10 && bluePct < 10) which correctly identifies "mainly red" pixels.

Unit 2 Progress Check: MCQ Part A

18. In the following code segment, str1 and str2 are String objects. The code segment is intended to print true if str1 and str2 are non-null strings that contain the same sequence of characters. The code segment is intended to print false otherwise.

```
boolean result = false;          // line 1
if(str1 != null && str2 != null)  // line 2
{
  result = str1 == str2;         // line 4
}
System.out.println(result);
```

Which of the following best explains the error, if any, in the code segment?

(A) In line 1, result should be initialized to true.

(B) In line 2, str1 and str2 should be compared to null with the equals method.

(C) In line 4, str1 and str2 should be compared to each other with the equals method. ✓

(D) The code segment contains no errors and works as intended.

Answer C

Correct. In line 2, the String objects are correctly compared with null using the != operator. Using == in line 4 checks whether str1 and str2 refer to the same String object. To check whether they have the same value, they should be compared using the equals method.